

Fly GACA — Project Roadmap

SECTION 00-strategy TYPE plan STATUS **active** OWNER Founder UPDATED 2026-06-16

The master plan from the end of Phase 0 to a shipped product, across all eleven phases. Companion to `phase0.md` (the detailed Phase 0 tracker). This file is a living checklist — edit the **Status** fields and tick the checkboxes as you go.

Last updated: 2026-05-24 **Status values:** `Not started` · `In progress` · `Blocked` · `Done`

Staleness check (2026-06-16): phase statuses below have not been re-verified since the date above. Treat as point-in-time. **TODO(owner):** re-confirm and bump "Last updated".

Sequenced by dependency, not by date. Solo build, no fixed deadline. The order below is the order things *can* be done in; where two tracks can run in parallel, that is called out explicitly.

Not legal advice. The legal items below depend on the Saudi IP lawyer's opinion (see `office/lawyer-brief.md`) and, for PDPL, on proper professional review.

Where things stand (24 May 2026)

Phase	Title	Status
0	Foundations	In progress — technical track done; legal track open
1	The library goes live	Done — live; custom domain + CI still pending
2	Captain Adel answers	Done — RAG function live on <code>gemini-2.5-flash</code> ; App Check + evals pending
3	Pilot accounts & tools	In progress — accounts system built, verified & live (Auth provider on, rules deployed); PDPL DPIA is the one gate left before inviting users
4	Arabic, offline & polish	In progress — bilingual engine + RTL live; homepage, About & all 4 section hubs translated; remaining inner pages + native review pending
5	Money & flight schools	Not started
6	Reach	Not started
7	The training platform	In progress — explanations, Captain Adel in quizzes, 162-question bank, exam-readiness analytics shipped
8	The library as a platform	Done — all 6 features shipped: study content, accident lessons, whole-library search, cross-links, reading paths, FAA comparison

Phase	Title	Status
9	Launch & visibility	In progress — site deployed & live; SEO/structured-data/analytics done; custom domain + go-to-market pending
10	Captain Adel, production-grade	In progress — rate limits, input guards & eval harness shipped; App Check, CI gate & budget alerts pending
11	Depth across the practical sections	Done — Tools 17→21, Guides 7→11, Study bank 60→162 + flashcards mode

The road ahead, in priority order: the build is far ahead of the paperwork. The remaining work is mostly **gates and finishing**, not new construction — (1) the **legal track** (Phase 0): engage the Saudi lawyer, lock the name, register the entity; (2) the **PDPL DPIA**, which is the one gate before pilot accounts can be opened to the public; (3) **finish Phase 4** — translate the remaining inner pages and get a native Arabic review; (4) the small **console chores** — custom domain, App Check, budget alerts, the analytics token, a CI deploy gate; then (5) **Phase 5** (monetisation, gated by the legal entity) and (6) **Phase 6** (native apps + public API). The legal/entity track is now the critical path — the product itself is built.

Phase 0's technical track (repo, Firebase, VPS) is **done**; what remains in Phase 0 is the legal track — engaging the lawyer (P0-1/P0-2) and registering the entity (P0-3). **Phases 1, 2, 8 and 11 have shipped**: the library, the in-app reader, the 21 flight tools, the 11 guides and the Study section (Ground School, 12 quiz banks, flashcards and a timed mock exam) are all live at `flygaca-firebase.web.app`, Captain Adel answers with cited sources, and the library now powers safety lessons, reading paths and whole-library search. Phase 7 (the training platform) is largely shipped and **Phase 9 — the public launch — is the immediate next step**.

The critical path — what gates what

The single most important point: Phase 1 is not blocked. Because Phase 1 was decided to launch as a **deep-link index** (it links to GACA's own copies rather than rehosting the texts) and has **no user accounts**, the Phase 1 build can start *now* and run in parallel with the legal track. The lawyer's opinion gates the *corpus rehosting upgrade* and the *final brand name* — not the Phase 1 build itself.

The gating chain for everything else:

- **Lawyer opinion (P0-1)** → unlocks rehosting the actual document texts (a Phase 1 upgrade) and confirms the AI-corpus position for Phase 2.
- **Lawyer opinion + name decision (P0-2)** → locks the final brand name → unlocks **P0-3** (entity registration needs the name) and final branding everywhere.
- **Legal entity (P0-3)** → unlocks the `.sa` domain, a business bank account, and the payment gateway + VAT in Phase 5.
- **Blaze billing plan** → required before **Phase 2** (Cloud Functions need it).
- **PDPL DPIA** → required before **Phase 3** (user accounts and personal data).

Recommended parallelism: run the legal track (finish Phase 0) and the **Phase 1 build** at the same time. Do not wait idle for the lawyer.

Phase 0 — Foundations (closing out)

Goal: legal, corporate and technical groundwork. **Status:** In progress. **Detailed tracker:**

`phase0.md` .

Legal — engage the lawyer (P0-1 & P0-2)

- ☐ Pick 2–3 firms from `office/lawyer-shortlist.md`
- ☐ Send `office/lawyer-brief.docx` ; request a fixed-fee quote, turnaround, and a conflict check (do they act for GACA?)
- ☐ Compare quotes; engage one firm
- ☐ Receive the written opinion on redistribution rights and the name
- ☐ Record the answers in `phase0.md` — host vs. deep-link; the name verdict

Legal — lock the name (P0-2, after the opinion)

- ☐ Decide: keep "Fly GACA", demote it to a tagline, or rebrand
- ☐ Lock the final legal/brand name (this feeds P0-3)
- ☐ Finalise `legal/DISCLAIMER.md` with the lawyer-approved wording
- ☐ If the name changes: update the repo, pages, brand assets and domains accordingly

Legal entity (P0-3, after the name is locked)

- ☐ Choose the legal structure (sole proprietorship vs. LLC)
- ☐ Register the Commercial Registration on the Saudi Business Center
- ☐ Register with Monshaat (SME)
- ☐ Check NTDP eligibility and apply
- ☐ Open a business bank account
- ☐ Record the CR number and entity details in `phase0.md`

Housekeeping

- ☐ Delete the duplicate Firebase project `fly-gaca-495116`
- ☐ Reconcile the old `FLyGACA/Library` repo — archive it, or fold it into `library/`
- ☐ Verify the data-residency of Gemini / Vertex AI inference (needed before Phase 2)

Exit criteria: the redistribution-rights question has a documented, lawyer-confirmed answer; the name is locked; the legal entity is registered with a business bank account.

Phase 1 — The library goes live

Goal: a fast, public GACAR library at flygaca.com — no login, no AI. Static PWA, client-side search, legal pages, deployed to Firebase Hosting. Launches as a **deep-link index** (links to GACA's copies); rehosting the texts is a later upgrade gated on the lawyer. **Status:** Not started — **can start now**, in parallel with the Phase 0 legal track. **Gated by:** nothing for the index build · **Gates:** Phase 2.

Content & corpus index

- ☒ Inventory the corpus — 74 GACAR parts, 61 aerodromes (AIP AD) and 13 VFR charts
- ☒ Build the index data files — gacar / aerodromes / charts indexes in `assets/data/`
- ☒ Define the search index format — client-side JSON, searched in `assets/js/library.js`
- ☒ In-app reader — 74 GACAR parts rendered on the Ivory reading surface from the corpus (`assets/data/parts/`, shown by `document.html`). Note: hosting the regulation text itself still needs the lawyer's confirmation under P0-1.

Design system & build (vanilla JS PWA, no framework)

- ☒ Implement the brand as CSS — Falcon palette tokens + Cairo type, dark-first, RTL-aware (`assets/css/tokens.css`, `base.css`); brand assets in `assets/`
- ☒ Build `flygaca.html` — landing page: hero, features, Captain Adel waitlist, footer
- ☒ Build `library.html` — three-section library (Regulations · Aerodromes · VFR Charts) with search
- ☒ Build the client-side search — title search, plus full-text search across all 8,256 GACAR sections, with snippets that deep-link into the reader
- ☒ Build the document detail view — `document.html`: regulation reader, aerodrome dashboard, and a zoomable VFR chart viewer; plus the interactive map (`map.html`)
- ☒ PWA shell — `site.webmanifest`, app icons, and an offline service worker (`sw.js`)
- ☐ Responsive layout; confirm the unofficial-status disclaimer renders on every page
- ☐ Accessibility baseline (semantic HTML, keyboard nav, contrast)

Legal & policy pages (business track, with Phase 1)

- ☒ Disclaimer page — `disclaimer.html` (draft — pending lawyer review before launch)
- ☒ Terms of Use — `terms.html` (draft — pending lawyer review before launch)
- ☒ Privacy Notice — `privacy.html`, PDPL-aware (draft — pending lawyer review + DPIA)
- ☐ Attributions / sources page (credit GACA as the authoritative source)
- ☐ Cookie / analytics notice if analytics are used

Deploy

- ☒ Configure `firebase.json` + `.firebaserc` (hosting, root rewrite, caching headers)
- ☐ Run the first deploy — `firebase deploy --only hosting` (see `office/runbook-deploy.md`)

- ☐ Connect the custom domain flygaca.com (DNS) — automatic SSL
- ☐ GitHub Actions: build + deploy to Firebase Hosting on push to `main`
- ☐ Launch smoke test on production

Go-to-market (business track)

- ☐ Wire the waitlist capture (store emails in Firestore me-central2)
- ☐ Publish the first regulatory explainers from the Captain Adel LinkedIn
- ☐ Seed flight schools and pilot communities; start collecting the waitlist

Exit criteria: flygaca.com serves a fast public GACAR index with working search and legal pages, deployed through CI, with a live waitlist.

Phase 2 — Captain Adel answers

Goal: the AI study assistant — ingestion pipeline, vector store, a RAG Cloud Function, a bilingual cited chat UI, an eval harness, and App Check + rate limits. **Status:** Not started. **Gated by:** Blaze plan; Gemini/Vertex residency check; the lawyer's AI-corpus view is helpful here · **Gates:** the paywall in Phase 5.

Prerequisites

- ☐ Upgrade Firebase to the **Blaze plan** (requires a payment card) — Cloud Functions need it
- ☐ Confirm Gemini / Vertex AI inference data-residency is acceptable for the data flow
- ☐ Choose the vector store (e.g. Firestore vector search, or an index built on the VPS and served from a Cloud Function)

Ingestion pipeline (Python 3.11, on the VPS — public data only)

- ☐ Build PDF ingestion: parse GACAR / AIP / circular documents → clean text
- ☐ Implement chunking (`assistant/rag.py` , `functions/rag/`)
- ☐ Generate embeddings
- ☐ Build and store the vector index
- ☐ Tag every chunk with its effective date / AIRAC cycle (AIP freshness)
- ☐ Confirm the pipeline touches public data only (PDPL boundary)

RAG service & assistant

- ☐ Build the RAG Cloud Function in me-central2 (`functions/rag/`)
- ☐ Retrieval: query → embed → retrieve → rank → assemble context
- ☐ Integrate Gemini with the Captain Adel system prompt (`assistant/`)
- ☐ Write the conservative, cited system prompt — every answer cites sources, never implies official status, and adds the AIP "not for operational use" line where relevant

- ☐ Enforce the corpus policy — host-safe + original material only; cite-only for ICAO Annexes and POH/AFM, never reproduced
- ☐ Bilingual answering (English and Arabic)

Chat UI

- ☒ Build `chat.html` — chat interface with Captain Adel's avatar, wired to a configurable RAG endpoint; assistant scaffold + system prompt in `assistant/`, BM25 corpus built at `functions/rag/_chunks.json.gz`
- ☐ Streaming responses with inline citations
- ☐ Clear messaging for rate limits and refusals

Security & abuse control

- ☐ Register **App Check**; enforce it on the RAG function
- ☐ Rate limits (per IP / per session, pre-accounts)
- ☐ Basic abuse and prompt-injection guards

Evaluation

- ☐ Build the eval harness (`evals/`) with authored question sets
- ☐ Evals for answer accuracy, citation correctness, and correct refusals
- ☐ Run evals; iterate on the prompt and retrieval until the bar is met
- ☐ Wire a regression eval into CI

Exit criteria: Captain Adel answers aviation questions with correct citations and disclaimers, behind App Check and rate limits, passing the eval bar.

Phase 3 — Pilot accounts & tools

Goal: Firebase Authentication, a Firestore data model with strict rules, a digital logbook with currency tracking, calculators, and PDPL data rights (export + deletion). **Status:** In progress — the full accounts system is **built and code-verified**; it must not be switched on for the public until the PDPL DPIA is signed off. **Gated by (launch, not build):** the PDPL DPIA.

Prerequisite — PDPL compliance (business/legal track)

- ☐ Complete the PDPL compliance program and the Data Protection Impact Assessment (DPIA)
- ☐ Identify the data controller; register / appoint as PDPL requires
- ☐ Update the Privacy Notice to cover accounts, the logbook, and chat history
- ☐ Confirm all personal data stays in the Kingdom (Firestore me-central2 / Cloud Functions); the VPS stays public-data-only

Authentication

- ☒ Sign-up / sign-in / password-reset UI — `account.html` + `assets/js/auth.js`, Firebase Auth (email/password) wrapped as a shared ES module
- ☒ `settings.html` — profile, change-password (with re-authentication), and the account/identity panel
- ☒ Email/Password provider enabled in the Firebase console; sign-up tested live
- ☐ Email verification

Data model & security rules

- ☒ Firestore data model designed and implemented — `users/{uid}` profile + `users/{uid}/logbook/{entry}`, documented in `assets/js/store.js`
- ☒ Strict security rules — `firestore.rules`: per-user isolation, no cross-user reads, field validation on writes
- ☒ Security rules deployed (`firebase deploy --only firestore:rules`) — live and verified: a test sign-up wrote a profile that passed the `isOwner` rule
- ☐ Add emulator-based automated rule tests

Digital logbook

- ☒ Logbook entry UI — `logbook.html` + `assets/js/logbook.js`: add / edit / delete flights, all fields (hours, landings, approaches, route, remarks), running totals
- ☒ Currency tracking — `dashboard.js` computes day/night passenger recency, IFR recency, and medical / flight-review expiry from the logbook
- ☒ Currency dashboard — `dashboard.html`: currency cards, logbook totals, recent flights

Calculators

- ☒ 21 aviation calculators and tools shipped in the Tools section (E6B, weight & balance, fuel, performance, conversions and more) — see Phase 11

PDPL data-subject rights

- ☒ Data export — `settings.html` downloads the full profile + logbook as JSON
- ☒ Account deletion — full erasure of all Firestore data, then the Auth account
- ☐ Consent management and a record of consent — pairs with the DPIA work

Exit criteria: pilots can register, keep a logbook with currency tracking, and use the calculators; PDPL data-subject rights work end to end; the DPIA is signed off and the Email/Password provider is enabled.

Phase 4 — Arabic, offline & polish

Goal: full Arabic RTL parity, an offline service worker, accessibility and Web Vitals, a Playwright E2E suite, and an error beacon. **Status:** In progress — the bilingual foundation (i18n engine, RTL, translated chrome

- homepage) has shipped; per-page Arabic content is the remaining bulk. **Gated by:** Phases 1–3 (the surfaces being polished).

Arabic & RTL

- ☒ Bilingual engine — `landing.js` carries an EN/AR engine on every page: persists the choice, sets `dir / lang`, translates the nav + footer chrome and the two shared legal blocks from a built-in dictionary, applies `data-en / data-ar` attributes, and keeps the brand wordmark LTR
- ☒ RTL layout — the whole CSS design system is built on logical properties, so `dir="rtl"` mirrors every page automatically; verified across the site
- ☒ Fully translated to Arabic — the entire main navigable surface: the homepage, About, all four section hubs (Library, Tools, Study, Guides), the Captain Adel chat page, the Safety section and Reading Paths, plus the 7 original guide pages. The engine handles plain text (`data-ar`), rich blocks with inline markup (`data-ar-html`) and form placeholders (`data-ar-ph`)
- ☐ Long-tail translation — the 21 individual tool/calculator pages, the study sub-pages, the account area, and the 4 newer guides
- ☐ Legal pages (disclaimer / terms / privacy) — translate **with** the lawyer review, not before; first-pass machine Arabic of legal text is inadvisable
- ☐ Arabic content QA with a native reviewer (aviation terminology accuracy) — the current Arabic copy is a first pass and is flagged for native review

Offline

- ☒ Offline service worker — `sw.js` precaches the app shell and runtime-caches visited content, with an offline navigation fallback
- ☐ Extend offline caching to library reader content
- ☐ Graceful offline UX (clear states, no broken flows)

Quality

- ☒ Accessibility baseline — semantic HTML, skip links, keyboard nav, an accessibility/mobile pass (Phase 1 close-out)
- ☒ Client-side error beacon — shipped in `assets/js/analytics.js` (Phase 9)
- ☐ Full accessibility audit — WCAG, screen-reader, contrast across the palette
- ☐ Web Vitals — set a performance budget; optimise LCP / CLS / INP; trim assets
- ☒ Playwright E2E suite (`tests/`) — `smoke.spec.js` loads all 43 public pages and fails on any JS error; `critical-flows.spec.js` exercises navigation, the bilingual engine, flashcards, the quiz and the chat UI. 50 tests, all passing. Run with `npm test` (see `tests/README.md`).

- ☐ Run E2E in CI on every push
- ☐ Cross-browser and cross-device QA

Exit criteria: full English/Arabic parity, works offline, passes the accessibility and Web Vitals bar, with a green E2E suite.

Phase 5 — Money & flight schools

Goal: turn Fly GACA into a revenue business — a free funnel, a paid Pro tier, a one-time Services line, a B2B flight-school product, Saudi payments wired to ZATCA, and the paywall to enforce it.

Status: Not started — **plan locked, prices decided** (below). **Gated by:** the legal entity (P0-3) for live payments; the build of the pages and paywall is not gated.

Guiding rule — never paywall the regulations

The GACAR library is public regulatory law: it stays free forever. It is the SEO funnel, the trust-builder, and the safest legal posture. Fly GACA charges for the *tools, teaching and AI* that act on the regulations — not for reading the rules.

Free tier — the funnel (no account required)

- The whole Library — 74 GACAR Parts, 21 handbooks, 61 aerodromes, 13 charts, 190 reference docs, full-text search, the in-app reader
- All 11 Guides, the Safety section, Reading Paths
- Captain Adel — **5 cited questions per month**, as a taste (this cap is both the conversion engine and the per-message cost cap; a monthly allowance is *lower* cost per free user than the old per-day cap, and a sharper paywall). Server window is `ADEL_FREE_PERIOD=month` (or `day`) with `ADEL_DAILY_FREE` as the count
- The three **funnel tools** (below) and the basic airport lookup — free by design

Pro tier — the individual paid product

- All 21 Flight Tools / calculators (tool *pages* stay public and indexable for SEO; the gate is on *interactive use*). Free users get a small trial allowance (≈ 3 runs/day) so they hit value before the wall
- **Unlimited Captain Adel** (fair-use soft cap; the Phase 10 rate limiter guards abuse)
- **Saudi Airport Directory** — every OE** airport: runways, frequencies, elevation, lighting, fuel/customs/handling, AIP cross-reference, searchable; static data from the official AIP
- **Live weather** — METAR/TAF for Saudi airports, from NOAA's aviationweather.gov feed (free, legal); pairs with the airport directory
- The full Study system — 12 quiz banks, the timed mock exam, exam-readiness analytics, the 34-lesson Ground School, flashcards
- The digital logbook + currency tracking + cross-device sync

- Billed two ways: a **monthly / annual subscription**, and a one-time fixed-term **Exam Term** (120 days) for the pass-and-leave student

Services — one-time Prep Packs (the new product line)

Fly GACA is independent of GACA, so it cannot sell a licence conversion or a test — only GACA does that. What it sells is **preparation**: structured playbooks, document checklists, process roadmaps and cited study/mock material that get a candidate ready. Paid once.

- **License Conversion Prep Pack** — for foreign ATPL/CPL/PPL holders converting to a GACA licence: the conversion-path roadmap, the full document checklist, a GACA air-law conversion study pack built only from the GACAR, mock questions, and a "what to expect" process walkthrough. Includes 90 days of Pro + Captain Adel Conversion Mode
- **ELPT / SAELPT Prep Pack** — English Language Proficiency prep against the six ICAO descriptors: phraseology drills, practice scenarios, a mock-interview structure, a self-scoring rubric. Includes 30 days of Pro
- **AIP Prep Pack** — how to actually use the Saudi AIP for exams and checkrides: structure, how to find what an examiner asks, quizzes from real navigation questions. Includes 30 days of Pro
- **Conversion Bundle** — all three Prep Packs together
- **Captain Adel 1:1 Consult** — a live 180-minute deep-dive session; limited slots per week (it spends the founder's real time)
- **Conversion Pack Premium** — the License Conversion Prep Pack + one 1:1 consult + 6 months of Pro

Fees not included. Every pack price covers Fly GACA's prep material and guidance *only*. All official costs — GACA conversion/application fees, the air-law exam fee, the ELPT/SAELPT test fee, the medical — are paid by the candidate directly to GACA or the testing body. Stated in plain language on the pricing page and at the top of every pack. Pack content is built only from official GACAR / AIP / GACA-published material — no fabricated exam specifications.

The funnel — how the tools guide interest

The free tools are lead magnets that diagnose a need and hand the user to the matching paid offer:

- **Conversion Eligibility Checker** (free) → state / licence / hours / ratings → an instant read on the likely GACA conversion path and requirements → CTA to the License Conversion Prep Pack
- **ELP Readiness Self-Check** (free) → self-assessment against the ICAO descriptors → estimated band + weak areas → CTA to the ELPT Prep Pack
- **AIP Quiz** (free) → quick quiz answerable from the Saudi AIP → score + weak sections → CTA to the AIP Prep Pack
- **Basic airport lookup** (free) → upsells the full Airport Directory + weather in Pro

The other ~18 calculators sit behind Pro with a small free trial allowance, and each paywall upsells contextually. Every Prep Pack bundles a stretch of Pro so the buyer experiences the calculators and Captain Adel — services feed the subscription rather than competing with it.

Captain Adel — tiering & economics

- Free: 5 cited questions/month — the most magnetic feature on the site and the single best conversion point; a monthly cap controls per-message API cost while making the paywall bite sooner than a daily reset would
- Pro: unlimited (fair-use soft cap; the Phase 10 rate limiter guards abuse). At the Pro price a heavy user's token cost is a few riyals — healthy margin
- Inside the Prep Packs: **Captain Adel Conversion Mode** — a focused assistant primed on the conversion / ELP / AIP material, included with each pack

Fly GACA for Schools — the B2B revenue engine

- Seat-based bulk licensing — a school buys N cadet seats, each cadet gets Pro
- A school **admin dashboard** — cohort exam-readiness, per-cadet progress, who is falling behind (a genuine instructor tool, not just billing)
- Annual contracts, ZATCA-compliant invoicing, per-seat price below individual Pro

Price sheet (SAR) — decided

The full derivation (market benchmarks, anchoring logic, the phased revenue plan) lives in `office/monetization.md` — that document is the source of truth; `pricing.html`, `schools.html` and `flygaca.html` must match it.

Subscriptions

- Pro Monthly — **59**
- Pro Annual — **349** ($\approx 29/\text{mo}$ billed yearly, ~6 months free vs monthly), with a 7-day free trial
- Exam Term — **199** (120 days full access, one-time)

Pricing note. Monthly sits at 59 and annual at 349 deliberately: the wide spread makes the annual the obvious buy (the decoy/anchor effect), pulls cash forward, and reduces churn. The annual headline is framed as **~SAR 29/month** on the marketing surfaces. Founding-member waitlist offer: annual locked at **299** for year 1 (first 500 members).

Services — one-time Prep Packs (*Fly GACA prep only — official fees not included*)

- License Conversion Prep Pack — **899**, launch offer **699** (incl. 90 days Pro)
- ELPT / SAELPT Prep Pack — **349** (incl. 30 days Pro)
- AIP Prep Pack — **299** (incl. 30 days Pro)
- Conversion Bundle (all three) — **1,299** (saves 248)

Premium — 1:1

- Captain Adel 1:1 Consult — **899 / 180-min session**, with a **699 launch offer** (founding rate, limited slots)
- Conversion Pack Premium — **1,699** (pack + one 1:1 consult + 6 months Pro)

Schools (B2B) — minimum 10 seats; see `docs/b2b-pipeline.md` for the sales motion

- **299 / seat / year** — **249** at 25+ seats, **199** at 75+ seats; admin dashboard included
- Founding-partner rate for the first 2–3 schools: **199** flat, year 1

Build track — what can be built now (gateway deferred)

- ☐ Subscription / entitlement data model — `users/{uid}` carries a plan (`free` | `pro` | `school`), one-time pack entitlements, term and expiry; security rules enforce it
- ☐ Paywall + gating layer — gate interactive tool use (≈ 3 /day trial), the full Study system, the logbook, the airport directory/weather and unlimited Captain Adel; the 5/month Captain Adel free counter; a clean contextual upgrade prompt at each limit
- ☐ The three funnel tools — Conversion Eligibility Checker, ELP Readiness Self-Check, AIP Quiz — each free, each with a CTA to its pack
- ☐ Saudi Airport Directory + live METAR/TAF weather
- ☐ The three Prep Packs — content + delivery
- ☐ `pricing.html` — public pricing page: Free vs Pro vs Services vs Schools
- ☐ `schools.html` — the B2B offering and a "request seats" enquiry path
- ☐ School admin dashboard — seat management + cohort analytics
- ☐ Subscription / purchase management UI — upgrade, cancel, billing history, pack purchases (checkout stubbed until the gateway is live)

Business track — gated on the legal entity (P0-3)

- ☐ ZATCA VAT registration
- ☐ ZATCA Fatoora e-invoicing onboarding (integration + compliance)
- ☐ Choose a mada-capable payment gateway (Moyasar / HyperPay / PayTabs / Tap)
- ☐ Integrate the gateway via Cloud Functions; recurring subscription billing + one-time pack/consult payments
- ☐ Wire ZATCA Fatoora — issue compliant e-invoices
- ☐ Bulk billing and invoicing for schools

Exit criteria: individuals can subscribe, buy an Exam Pass, or buy a one-time Prep Pack and pay; ZATCA-compliant invoices are issued; flight schools can buy and manage cadet seats; the free library remains open to everyone.

Phase 6 — Reach

Goal: a Capacitor native wrapper for the app stores, and a documented public API. **Status:** Not started. **Gated by:** a stable, polished product (Phases 1–5).

Native apps

- ☐ Wrap the PWA with Capacitor
- ☐ iOS build — Apple Developer account, App Store listing, submission and review
- ☐ Android build — Google Play listing, submission and review
- ☐ Store assets — screenshots, descriptions, privacy disclosures, compliance

Public API

- ☐ Design the public API surface (read access to the library / regulations)
- ☐ API authentication (keys), rate limits, and quotas
- ☐ Write and publish the API documentation
- ☐ Developer onboarding

Exit criteria: native apps are live in the App Store and Google Play; a documented public API is available.

Phase 7 — The training platform

Goal: turn the Study section from a quiz into a genuine exam-prep system, tighten the practical tools, and play to Fly GACA's two unfair advantages — it is the only platform built on **GACAR**, and Captain Adel is an AI instructor grounded in the cited regulation text. Drawn from a competitive scan of Gleim, Sporty's, King Schools, SkyVector and ForeFlight. **Status:** In progress — training depth shipped; per-exam profiles, stage checks and the cross-device foundation remain. **Gated by:** nothing for the training items; the cross-device items want Phase 3 accounts.

Training & test-prep depth (priority)

- ☒ Expand the question bank — 162 questions across 12 banks (VFR, airspace, AIP/AIS, radio/ELPT, air law, licensing, medical, aircraft/equipment, weather, aerodynamics, human factors, navigation), each tagged to its GACAR Part / ICAO subject
- ☒ Add an explanation + GACAR citation to **every** question, deep-linked into the `document.html` reader — the feature no FAA-based competitor can match
- ☐ Realistic per-exam mock profiles — real question count, time limit, pass mark and subject weighting for each GACA written exam (one 25-question timed mock is live; per-path PPL/CPL/ATPL profiles still to build)
- ☒ Weak-area analytics + an "exam readiness" score (`study-progress.js`); review mode re-runs missed questions (`quiz.html?review=1`)

- ☒ Wire Captain Adel into the study flow — every quiz answer carries an "ask Captain Adel" hand-off, and Ground School lessons open straight into him
- ☐ Stage checks between Ground School modules + an end-of-course completion record (an unofficial personal study record, not a GACA endorsement)

Practical tools

- ☐ Live METAR/TAF lookup for Saudi aerodromes (OE** ICAO) from a public aviation-weather feed — surfaced on the aerodrome pages, kept separate from Captain Adel
- ☐ Interactive pan/zoom/measure VFR chart viewer with the 61 aerodromes overlaid
- ☐ Route-planner depth — a leg-by-leg heading/distance/time/fuel table, plotted on a map

Foundation

- ☐ User accounts + cloud sync (Phase 3) so progress, bookmarks, saved plans and quiz history follow the user across devices — the base every analytics feature above needs

Immediate focus — the first three to build: the deep-linked explanations, Captain Adel in the study flow, and expanding the question bank.

Exit criteria: the Study section works as a per-exam prep system — a large tagged question bank, every question explained and cited, realistic mock exams, and progress analytics that tell a candidate when they are ready.

Phase 8 — The library as a platform

Goal: turn the integrated library — 74 GACAR Parts, 21 handbooks, 61 aerodromes, 13 charts and 190 reference documents, all in Captain Adel's corpus — from a reference shelf into the engine behind study content, safety teaching and connected learning. **Status: Done** — all six features shipped. **Gated by:** nothing — the corpus and reader were in place.

Content generation

- ☒ Auto-generated cited study content from the corpus — the question bank grew from 60 to 162 questions across 12 banks, every item explained and cited; flashcard decks draw from the same bank (`study/flashcards.html`)

New learning surfaces

- ☒ "Learn from accidents" — `safety.html` , a safety section built on 54 NTSB accident lessons, each one openable into Captain Adel to ask how the GACAR guards against it
- ☒ GACAR ↔ FAA side-by-side — the `faaComparePanel()` on GACAR Part readers pairs the Saudi rule with its FAA equivalent (valuable for foreign-licence conversion)
- ☒ Curated reading paths — `paths.html` , 6 guided document tracks across 38 steps (private pilot, drone operations, the Class 1 medical and more)

Connecting it up

- ☒ Related-reading cross-links — `addRelated()` surfaces the closest handbook chapter, advisory circular, ICAO annex and FAA section on every reader page (TF-IDF similarity, `related-index.json`)
- ☒ Whole-library full-text search — `renderLibSearch()` searches a 46,036-entry index (`library-search.json`) spanning the GACAR, the 21 handbooks and the 190 reference docs

Outcome: the library now produces study material and powers connected, cross-referenced learning — the NTSB "Learn from accidents" section is the most distinctive piece, as no competitor pairs Saudi regulation with real accident lessons.

Exit criteria: the library actively produces study material and powers connected, cross-referenced learning — not just a place to read documents.

Phase 9 — Launch & visibility

Goal: take Fly GACA from "built" to "launched" — the whole platform live on the real domain, findable, measured and announced. The product is feature-complete; this phase puts it in front of pilots. **Status:** In progress — the findability track (SEO, structured data, sitemap, analytics) is done; the deploy and the domain remain. **Gated by:** nothing technical; the legal pages depend on the lawyer review.

Deploy & domain

- ☒ Deployed the full current stack — hosting + the Captain Adel Cloud Function — live at `flygaca-firebase.web.app`
- ☐ Connect the custom domain `flygaca.com` — DNS records, automatic SSL
- ☐ Confirm the GitHub repo is fully pushed; add a CI deploy on push to `main`

Findability

- ☒ SEO pass — every public page carries a unique title and description, Open Graph + Twitter cards (absolute image URLs) and a canonical link; the 7 unfinished stub pages are `noindex`; sitewide Organization + WebSite JSON-LD on the homepage and Article + Breadcrumb + FAQ JSON-LD across all 11 guides
- ☒ `sitemap.xml` rebuilt to all 47 current public pages; `robots.txt` confirmed current — **still to do:** submit the sitemap to Google Search Console / Bing after deploy
- ☒ Privacy-respecting analytics + a client-side error beacon — `assets/js/analytics.js` on every page: cookieless Cloudflare Web Analytics (dormant until a token is set, no consent banner needed) and an error beacon that logs uncaught errors to a localStorage ring buffer with an optional POST endpoint. Privacy Notice updated to disclose it. — **still to do:** create the Cloudflare Web Analytics site and paste the token

Go to market

- ☐ Finalise the legal pages once the Saudi lawyer review lands
- ☐ Soft launch — flight schools, pilot communities, the Captain Adel LinkedIn
- ☐ Convert the waitlist into first users

Exit criteria: `flygaca.com` is live, indexed, measured, and reaching real pilots.

Phase 10 — Captain Adel, production-grade

Goal: harden the chat AI for real, untrusted traffic — abuse protection, cost control and a measured quality bar. These were Phase 2's exit criteria, deferred to ship the MVP; this phase closes them. **Status:** In progress — the code-side hardening (rate limits, input guards, eval harness) has shipped; the console-side items (App Check, CI gate, budget alerts) remain. **Gated by:** real traffic starting (Phase 9).

- ☐ Register **App Check**; enforce it on the `chat` Cloud Function — needs the Firebase console (register a reCAPTCHA provider, then verify the token in `index.js` and attach it in the browser). Steps documented in `office/runbook-captain-adel.md`.
- ☒ Rate limits — per IP and per session — `functions/rag/ratelimit.js`, a sliding-window limiter (per-IP, a 30 s burst window, and per browser session); a blocked turn returns HTTP 429 and `chat.js` shows a friendly message. Limits are tunable via `ADEL_RL_*` env vars.
- ☒ Prompt-injection and abuse guards — `functions/rag/guards.js`: control-character stripping, message/history length caps, and a soft injection detector that hardens a flagged turn (appends a security note to the system instruction) rather than producing false-positive blocks.
- ☒ Build the eval harness (`evals/`) — `cases.json` (17 cases across citation, refusal, injection and behaviour) + `run.js`, a scoring runner that exits non-zero on any failure. Runs live against the agent with a `GEMINI_API_KEY`.
- ☐ Wire a regression eval into CI — `node evals/run.js` as a GitHub Actions gate on any change to `functions/rag/` or the system prompt.
- ☐ Budget alerts on Firebase and the Gemini API; watch the 2 GiB cold-start cost — needs the console (steps documented in the runbook).

Exit criteria: Captain Adel is protected against abuse and runaway cost, and passes a documented accuracy / citation / refusal bar.

Phase 11 — Depth across the practical sections

Goal: make the three working sections a pilot uses every day — Tools, Guides and Study — wider and more usable, so each is a genuine destination rather than a thin shelf. A breadth pass, run across all three at once. **Status: Done** — the expansion below shipped together. **Gated by:**

nothing.

Tools — 17 → 21 calculators

- ☒ Top of Descent — descent point, rate and angle from cruise to a target altitude
- ☒ Fuel Planner — trip, reserve, alternate and taxi fuel with a total endurance figure
- ☒ Time–Speed–Distance — solve any of the three from the other two, with leg support
- ☒ Cloud Base — convergence of temperature and dew point into an estimated cloud base
- ☒ All four follow the self-contained `tools/density-altitude.html` pattern (calc-layout / calc-panel / calc-field / calc-out, inline script) and are wired into the Tools hub, the nav and the service-worker precache

Guides — 7 → 11 how-to guides

- ☒ Reading a METAR & TAF — decoding the standard aviation weather reports field by field
- ☒ Decoding NOTAMs — the contraction set and how to read a raw NOTAM
- ☒ The AIRAC Cycle — why aeronautical data changes every 28 days and how to stay current
- ☒ Airspace Explained — ICAO Classes A–G in plain language, mapped to the Saudi system
- ☒ All four match the Guides section styling and are registered in the hub, nav and SW

Study — exam prep, deeper

- ☒ Question bank grown 60 → 162 across 12 topic banks, every item explained and cited
- ☒ Flashcard review mode — `study/flashcards.html` + `assets/js/flashcards.js`: pick a deck (per-bank or a 40-card mixed deck), flip each card, mark "Got it" or "Review again", and the misses loop back for a second pass; linked from the Study hub
- ☒ Exam-readiness panel and missed-question review mode (carried from Phase 7)

Outcome: Tools and Guides are broad enough to cover a real cross-country workflow, and Study now offers three ways through the same bank — quizzes, a timed mock exam and flashcards.

Exit criteria (met): each of the three sections is wider and more usable; everything is wired into navigation and cached for offline use.

Cross-cutting — ongoing from Phase 1 onward

These are not one-off tasks; they run continuously once the relevant phase ships.

- ☐ **AIP / AIRAC freshness** — the AIP changes every 28-day AIRAC cycle. From Phase 1/2, a routine to refresh AIP-sourced content and effective dates.
- ☐ **Regulatory content freshness** — track GACAR amendments and re-ingest.
- ☐ **Security & dependencies** — keep Dependabot current; rotate keys; review rules.
- ☐ **Eval regression** — re-run the Captain Adel eval harness on prompt/model changes.

- ☐ **Go-to-market** — continuous LinkedIn explainers, community seeding, waitlist → user conversion, and flight-school outreach.
- ☐ **Backups & monitoring** — Firestore backups, uptime and error monitoring.

Phase 12 — The ecosystem (LTV & retention)

Goal: grow Fly GACA from a tools-and-answers product into the integrated ecosystem a Saudi pilot stays subscribed to for years — by deepening the AI, adding exclusive Pro features, and extending the B2B/instructor side. **Status:** Not started — sequenced below. **Guiding principle (founder's call):** do **not** build it all before launch. Ship the core paywalled product, then surface the rest as "**Coming soon to Pro**" to drive retention — a live, evolving platform is a reason not to cancel.

Already shipped — reuse, don't rebuild

Much of the "ecosystem" already exists and only needs to be marketed as such:

- **Smart digital logbook + GACAR currency/recency alerts** — `logbook.html`, `logbook.js`, `currency.js` (Part 61), `dashboard.html` (Pro-gated)
- **VFR/IFR planning + NavLog inputs** — the route-planner, flight-plan, W&B, fuel and TSD tools under `tools/`
- **Checkride mock exams / question banks** — `study/checkride.html`, `study/quiz.html`, `study/groundschool.html` and the tagged banks in `assets/data/`
- **School admin dashboard (concept) + bulk licensing** — `schools.html`, `functions/school.js`

New builds — ranked by ROI (biggest pain point first)

1. **AI Oral-Exam / Checkride Examiner** (*flagship — build first*) — a conversational AI that role-plays a GACA examiner: asks licence-specific (PPL/CPL/IR) questions grounded in the corpus, grades the answer and cites the exact Part/section. Highest willingness-to-pay moment, and the **lowest-cost** new AI feature because it reuses Captain Adel's RAG brain (`src/brain/` in `FlyGACA/Captain-Adel`) — a new `mode: 'examiner'` persona behind the `/v1/chat` contract, **eval-gated**; new gated page `study/oral-exam.html` reusing `chat.js`. Pairs with the Exam Term SKU. *No "guaranteed to pass" claims; always cite + verify against official source.*
2. **GACAR / AIP Change Digest** — scheduled monitor + AI summary of regulation/AIP changes into plain-language bullets, on the dashboard and as an opt-in digest, every item linked to the official source. Heaviest ops (monitoring + email).
3. **Instructor Dashboard** — assign custom mock exams, per-cadet weak-area analytics, review cadet W&B. Needs a new instructor↔cohort↔cadet Firestore model + `firestore.rules` coverage
 - `tests/rules.test.js`; extends `school.js`.
4. **Enterprise / white-label portal** — academy-branded curriculum + content management. Largest; future.

This pass (shipped now)

- ☒ **"Coming soon to Pro" retention shelf** — bilingual section on `pricing.html` listing the Oral-Exam Examiner, Change Digest and Instructor Dashboard (compliance-safe, no pass guarantees)
- ☒ This roadmap section

Exit criteria: the Oral-Exam Examiner ships eval-gated as the flagship Pro feature; the remaining items are sequenced with their data-model / ops prerequisites; the GACAR library and guides stay free throughout.

Decision & risk checkpoints

- **Lawyer opinion** — until it lands, do not rehost the corpus or finalise the name. Phase 1 ships as a deep-link index regardless.
 - **Name** — a rebrand verdict touches the repo, domains, brand assets and the entity name; cheaper to absorb before Phase 1's public launch than after.
 - **PDPL** — the DPIA is a hard gate for Phase 3; budget time for it.
 - **Costs that need a card** — the Blaze plan (Phase 2), the payment gateway and VAT (Phase 5), and the Apple/Google developer accounts (Phase 6).
 - **Solo-founder load** — Phases 2 and 3 are the largest; Arabic QA (Phase 4) and the ZATCA/payments integration (Phase 5) are the most likely points to bring in help.
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Living document — keep it in step with `phase0.md` and the repo. Not legal advice.